PROJECT REPORT

NUMPY STUDENT MARKS ANALYSIS

1. Introduction

Numerical analysis is an important part of data analysis and machine learning. Student marks can be summarized using statistical measures to understand overall performance and variation.

This project uses Python and NumPy to calculate descriptive statistics from a collection of student marks.

2. Objective

The objective is to create a NumPy array of student marks and calculate the mean, median, maximum, minimum, and standard deviation. The project also explains the meaning of the calculated results.

3. Problem Statement

Calculating statistical measures manually can become time-consuming when working with multiple observations.

This project addresses the problem by using NumPy's built-in functions to calculate the required statistics accurately and efficiently.

4. Tools and Technologies

Python 3 was used to write the program. NumPy was used to store numerical data and calculate the statistical measures. Visual Studio Code was used as the development environment, and GitHub was used to maintain the source code and documentation.

5. Dataset Description

The project uses manually created marks for ten students. Each mark is represented on a scale of 0 to 100.

The marks are:

78, 85, 92, 67, 88, 76, 95, 81, 72, 86.

This sample dataset is used to demonstrate the required calculations.

6. Methodology

First, the marks are stored in a NumPy array.

Next, the program validates the data to ensure that the array is not empty and the

marks are valid numbers between 0 and 100.

The following NumPy functions are then applied:

- `numpy.mean()` calculates the average.
- `numpy.median()` calculates the median.
- `numpy.max()` finds the highest mark.
- `numpy.min()` finds the lowest mark.
- `numpy.std()` calculates population standard deviation.

The results are displayed in the terminal along with a brief interpretation.

7. Results

Statistical Measure	Calculated Result
Number of students	10
Mean	82.00
Median	83.00
Maximum mark	95.00
Minimum mark	67.00
Population standard deviation	Approximately 8.05

8. Result Interpretation

The mean mark of 82 indicates a strong average performance in the sample.

The median mark of 83 is close to the mean, suggesting that the center of the distribution is similar under both measures.

The highest mark is 95, and the lowest mark is 67. The difference between them is 28 marks.

The population standard deviation is approximately 8.05 marks, indicating a relatively limited spread of marks around the mean for this dataset.

9. Learning Outcomes

This project provided practical experience with NumPy arrays and descriptive statistics. It demonstrated how built-in numerical functions can simplify data analysis and how validation and functions can improve code organization.

10. Conclusion

The NumPy Student Marks Analysis project successfully demonstrates the calculation of mean, median, maximum, minimum, and standard deviation using Python.

The results provide a simple summary of student performance and illustrate the usefulness of NumPy in numerical data processing.

11. Project Evidence

Insert a screenshot of the Python program's actual output here.

12. Project Links

GitHub Repository: https://github.com/venky1318/numpy-student-marks-analysis